44. Second quartile is same as ________.
45. Third quartile and ________ percentile are same.
46. Second quartile and ________ decile are equal
47. The decile which precedes 72th percentile is ________.
48. The percentage of values which lie between P_{10} and P_{56} is ________.
49. 25th percentile is same as ________ quartile.
50. Sixth decile is same as ________ percentile.
51. Percentage of values lying between first and sixth decile is ________.
52. Percentage of values which are greater than 66th percentile is ________.
53. In a frequency distribution fourth decile and ________ percentile are same.
54. The relation between second quantile Q_2, sixth decile D_6 and 80^{th} percentile is ________.
55. Median is same as ________ decile.
56. Median is same as ________ percentile.
57. Percentiles (quartiles, deciles) are always in ________ order.
58. The variate values which divide a series into five equal parts are called ________.
59. If in a series, 20 per cent values are greater than 75, then ________ = 75.
60. If in a discrete series 30 per cent values are less than 35, then ________ decile = 35.
61. If in a discrete series 75 per cent values are greater than 20, then ________ quartile = 20 or ________ percentile = 20.
62. The general name for quartile, quintiles, octiles, deciles and percentiles is ________.
63. The statement, "The mode is that value of a series which appears most frequently than any other", was given by ________.
64. The relation between first quartile and 25th percentile is ________.
65. The relation between 4th decile and 45th percentile is ________.
66. Percentage of values that lie between tenth and eighty fifth percentile is ________.
67. The number of types of statistical facts are ________ or ________.
68. For a symmetrical distribution, median in terms of quartiles is ________.
69. Geometric mean is ________ of the arithmetic mean of logarithmic values of a variable.
70. Harmonic mean is ________ of the arithmetic mean of the reciprocal of the observations of a data set.
71. Quadratic mean is preferred most, if the data set contains some ________ number.
72. If a variable takes some non-negative values $x_1, x_2, \ldots, x_n$, then the inequality that holds between H.M., G.M. and A.M. is ________.
73. Quadratic mean has ________ scale and units as that of original observation.
74. The application of quadratic mean is evidently found in ________.
75. The mean of 20 observations is 7. If each observation is multiplied by 3 and then 5 is added to it, then the mean of the new data set is ________.
76. The mean of a set of 10 observations is 4. Another set of 20 observations is added to it which makes the mean of the combined set equal to 6. The mean of second set is ________.
77. If $2y - 6x = 6$ and the mode of y is 66, then the mode of x is ________.
78. An average is that value of a distribution which represents ________.
79. No average can be regarded as ________ for all times and all data.
80. Mode is ________ defined.

SECTION-C

Multiple Choice Questions

Select the correct alternative out of given ones.

Q. 1 Mean is a measure of:
(a) location (central value)
(b) dispersion
(c) correlation
(d) none of the above

Q. 2 Which of the followings is a measure of central value?
(a) Median
(b) Standard deviation
(c) Mean deviation
(d) Quartile deviation

Q. 3 Which of the following represents median?
(a) First quartile
(b) Fiftieth percentile
(c) Sixth decile
(d) None of the above

Q. 4 If a constant value 50 is subtracted from each observation of a set, the mean of the set is:
(a) increased by 50
(b) decreased by 50
(c) is not affected
(d) zero

Q. 5 If a constant 5 is added to each observation of a set, the mean is:
(a) increased by 5
(b) decreased by 5
(c) 5 times the original mean
(d) not affected

Q. 6 If each observation of a set is multiplied by 10, the mean of the new set of observations:
(a) remains the same
(b) is ten times the original mean
(c) is one-tenth of the original mean
(d) is increased by 10

Q. 7 If each value of a series is multiplied by 10, the median of the coded values is:
(a) not affected
(b) 10 times the original median value
(c) one-tenth of the original median value
(d) increased by 10

Q. 8 If each value of a series is multiplied by 10, the mode of the coded values is:
(a) not affected
(b) one-tenth of the original modal value
(c) 10-times of the original modal value
(d) 100-times of the original modal value

Q. 9 If each observation of a set is divided by 2, then the mean of new values:
(a) is two times the original mean.
(b) is decreased by 2.
(c) is half of the original mean
(d) remains the same

Q. 10 Which of the following relations among the location parameters does not hold?
(a) Q_2 = median
(b) P_{50} = median
(c) D_5 = median
(d) D_6 = median

Q. 11 If the grouped data has open end classes, one cannot calculate:
(a) median
(b) mode
(c) mean
(d) quartiles

Q. 12 Geometric mean of two observations can be calculated only if:
(a) both the observations are positive
(b) one of the two observations is zero
(c) one of them is negative
(d) both of them are zero

Q. 13 Geometric mean is better than other means when,
(a) the data are positive as well as negative
(b) the data are in ratios or percentages
(c) the data are binary
(d) the data are on interval scale

Q. 14 Geometric mean is a good measure of central value if the data are:

(a) categorical
(b) on ordinal scale
(c) in ratios or proportions
(d) none of the above

Q. 15 Harmonic mean is better than other means if the data are for:
(a) speed or rates
(b) heights or lengths
(c) binary values like 0 and 1
(d) ratios or proportions

Q. 16 The correct relationship between A.M., G.M. and H.M. is:
(a) A.M. = G.M. = H.M.
(b) G.M. ≥ A.M. ≥ H.M.
(c) H.M. ≥ G.M. ≥ A.M.
(d) A.M. ≥ G.M. ≥ H.M.

Q. 17 Extreme value have no effect on:
(a) average
(b) median
(c) geometric mean
(d) harmonic mean

Q. 18 Geometric mean of two numbers $\frac{1}{16}$ and $\frac{4}{25}$ is:
(a) $\frac{1}{10}$
(b) $\frac{1}{100}$
(c) 10
(d) 100

Q. 19 Expenditure during first five months of a year is Rs. 96 per month and during last seven months is Rs. 120 per month. The average expenditure per month during whole year is:
(a) Rs. 108 per month
(b) Rs. 110 per month
(c) Rs. 100 per month
(d) Rs. 216 per month

Q. 20 Average strength of eleven members = 11.0. Average strength of the first six members = 10.5. Average strength of the last six members = 11.5.
The average strength of the sixth member is:
(a) 10.5
(b) 11.5
(c) 11.0
(d) 10.0

Q. 21 The average of the 7 number 7, 9, 12, x, 5, 4, 11 is 9. The missing number x is:
(a) 13
(b) 14
(c) 15
(d) 8

Q. 22 The mean proportion of 0.16 and 0.01 is:
(a) 0.4
(b) 0.17
(c) 0.085
(d) 0.04

Q. 23 A train covered the first 5 km of its journey at a speed of 30 km/h and next 15 km at a speed of 45 km/h. The average speed of the train was:
(a) 35 km/h
(b) 40 km/h
(c) 32 km/h
(d) 42 km/h

Q. 24 The second of the two samples has 50 item with mean 15. If the whole group has 150 items with mean 16, the mean of the first sample is:
(a) 18.0
(b) 15.5
(c) 16.5
(d) none of the above

Q. 25 For a group of 100 candidates, the mean was found to be 40. Later on it was discovered that a value 45 was misread as 54. The correct mean is:
(a) 40.50
(b) 39.85
(c) 39.80
(d) 39.91

Q. 26 A distribution consists of three groups having 40, 50 and 60 items with means 20, 26 and 15 respectively. The mean of the distribution is:
(a) 20
(b) 18
(c) 22
(d) 25

Q. 27 The average age of 50 students in a bus is 20 years. When the age of conductor is included, the average age is increased by one year. The age of the conductor is:
(a) 51
(b) 55
(c) 71
(d) 50

Q. 28 The average of five numbers is 40 and the average of another four numbers is 50. The average of all numbers taken together is:
(a) 44.44
(b) 45.00
(c) 45.55
(d) 90.00

Q. 29 The average temperature of two cities on first six days of a week is same. The temperature dropped in one city all of a sudden on the seventh day of the week. The average weekly temperature of the two cities differed by 0.5°C The difference between the six days average daily temperature of two cities and the seventh day temperature of the other city is:
(a) 2.5
(b) 3.0
(c) 3.5
(d) 2.0

Q. 30 What percentage of values is greater than 3rd quartile?
(a) 75 per cent
(b) 50 per cent
(c) 25 per cent
(d) 0 per cent

Q. 31 What percentage of values is less than 3rd decile?
(a) 30 per cent
(b) 70 per cent
(c) 40 per cent
(d) none of the above

Q. 32 What percentage of values lies between 5th and 25th percentiles.
(a) 15 per cent
(b) 30 per cent
(c) 75 per cent
(d) none of the above

Q. 33 There were 25 teachers in a school whose mean age was 30 years. A teacher retired at the age of 60 years and a new teacher was appointed in his place. The mean age of teachers in the school was reduced by one year. The age of the new teacher was:
(a) 25 years
(b) 30 years
(c) 35 years
(d) 40 years

Q. 34 If the A.M. of a set of two observations is 9 and its G.M. is 6. Then the H.M. of the set of observations is:
(a) 4
(b) $3\sqrt{6}$
(c) 3
(d) 1.5

Q. 35 The A.M. of two numbers is 6.5 and their G.M. is 6 The two numbers are:
(a) 9, 6
(b) 9, 5
(c) 7, 6
(d) 4, 9

Q. 36 If M_d, Q, D and P stand for median, quartile, decile and percentile respectively, then which of the following relation between them is true?
(a) $M_d = Q_2 = D_6 = P_{50}$
(b) $M_d = Q_3 = D_5 = P_{75}$
(c) $M_d = Q_2 = D_4 = P_{50}$
(d) $M_d = Q_2 = D_5 = P_{50}$

Q. 37 Which of the following relation is true between 3rd decile and 30th percentile?
(a) $D_3 = P_{70}$
(b) $D_3 = P_{50}$
(c) $D_3 = P_{30}$
(d) $D_7 = P_{30}$

Q. 38 Which of the deciles are less than first quartile?
(a) D_1 and D_3

(b) D_1 and D_2
(c) D_2 and D_3
(d) None of the above

Q. 39 In a factory there are 60 per cent labourers, 30 per cent scribes and 10 per cent executives. On average, the salary of a labourer is Rs. 1600 p.m, of a scribe Rs. 3000 p.m and that of an executive Rs. 8000 p.m. The average salary of a employee in the factory is:
(a) Rs. 4200 p.m
(b) Rs 1166 p.m
(c) Rs. 2660 p.m
(d) None of the three

Q. 40 The mean of seven observations is 8. A new observation 16 is added. The mean of eight observations is:
(a) 12
(b) 9
(c) 8
(d) 24

Q. 41 If the sum of N observations is 630 and their mean is 42, then the value of N is:
(a) 21
(b) 30
(c) 15
(d) 20

Q. 42 If the two observations are 10 and −10, then their harmonic mean is:
(a) 10
(b) 0
(c) 5
(d) ∞

Q. 43 If the observations are 5 and −5, their geometric mean is:
(a) 5
(b) −5
(c) 0
(d) none of the above

Q. 44 If the two observations are 20 and −20, their arithmetic mean is:
(a) 10
(b) 20
(c) 0
(d) none of the above

Q. 45 Can a quartile, a decile and a percentile be the median?
(a) Only quartile but not decile and percentile
(b) Quartile and decile but not percentile
(c) Decile and percentile but not quartile
(d) Quartile, decile and percentile, all the three

Q. 46 When all the observations are same, then the relation between A.M., G.M. and H.M. is:
(a) A.M. = G.M. = H.M.
(b) A.M. < G.M. < H.M.
(c) A.M. < G.M. < H.M.
(d) A.M. < G.M. < H.M.

Q. 47 The percentage of values used in case of 10 per cent trimmed mean is:
(a) 40 per cent
(b) 60 per cent
(c) 80 per cent
(d) 20 per cent

Q. 48 The average of $2n$ natural numbers from 1 to $2n$ is:
(a) $(n + 1)/2$
(b) $(2n + 1)/2$
(c) $n\,(n + 1)/2$
(d) $n\,(2n + 1)/2$

Q. 49 A man goes from his house to his office at the speed of 20 km/h and returns from his office to home at the speed of 30 km/h. His mean speed is:
(a) 24 km/h
(b) $10\sqrt{6}$ km/h
(c) 25 km/h
(d) none of the above

Q. 50 Mode is that value in a frequency distribution which possesses:
(a) minimum frequency
(b) maximum frequency
(c) frequency one
(d) none of the above

Q. 51 The value of the variable corresponding to the highest point of a frequency distribution curve represents:
(a) mean

(b) median
(c) mode
(d) none of the above

Q. 52 A frequency distribution having two modes is said to be:
(a) unimodal
(b) bimodal
(c) trimodal
(d) without mode

Q. 53 If modal value is not clear in a distribution, it can be ascertained by the method of:
(a) grouping
(b) guessing
(c) summarising
(d) trial and error

Q. 54 Shoe size of most of the people in India is No. 8. Which measure of central value does it represent?
(a) mean
(b) second quartile
(c) eighth decile
(d) mode

Q. 55 In a discrete series having $(2K + 1)$ observations, median is:
(a) K^{th} observation
(b) $(K + 1)^{th}$ observation
(c) $(K + 2)/2^{th}$ observation
(d) $(2K + 1)/2^{th}$ observation

Q. 56 The median of the variate values 11, 7, 6, 9, 12, 15, 19 is:
(a) 9
(b) 12
(c) 15
(d) 11

Q. 57 The median of the variate values 48, 35, 36, 40, 42, 54, 58, 60 is:
(a) 40
(b) 41
(c) 44
(d) 45

Q. 58 To find the median (mode), it is necessary to arrange the data in:
(a) ascending order
(b) descending order
(c) ascending or descending order
(d) any of the above

Q. 59 For a grouped data, the formula for median is based on:
(a) interpolation method
(b) extrapolation method
(c) trial and error method
(d) iterative method

Q. 60 Which of the measure of central tendency is not affected by extreme values?
(a) mode
(b) median
(c) sixth decile
(d) all the above

Q. 61 The middle value of an ordered series is called:
(a) 2nd quartile
(b) 5th decile
(c) 50 percentile
(d) all the above

Q. 62 Formula for directly calculating the mean $\overline{X}$ of an individual series is:

(a) $\overline{X} = \dfrac{\Sigma X}{N}$

(b) $\overline{X} = \dfrac{\Sigma fX}{N}$

(c) $\overline{X} = A + \dfrac{\Sigma\, dx}{n}$ where $dx = X - A$

(d) $\overline{X} = A + \dfrac{\Sigma\, dx}{N}$ where $dx = X - A$

Q. 63 Formula for calculating the mean of an individual series by short-cut method is:

(a) $\overline{X} = \dfrac{\Sigma X}{N}$

(b) $\overline{X} = \Sigma\, fX/N$

(c) $\overline{X} = A + \dfrac{\Sigma\, dx}{N}$

(d) $\overline{X} = A + \dfrac{\Sigma\, fdx}{N}$

Q. 64 Formula for calculating the mean of a discrete series by direct method is:

(a) $\bar{X} = \Sigma X/\Sigma f$

(b) $\bar{X} = \Sigma_i f_i X_i/\Sigma_i f_i$

(c) $\bar{X} = A + \Sigma_i f_i\, dx_i/\Sigma f_i$

(d) none of the above

Q. 65 Formula for calculating the mean of a discrete series by short-cut method is:

(a) $\bar{X} = \Sigma X/\Sigma f$

(b) $\bar{X} = \Sigma f_x/\Sigma f$

(c) $\bar{X} = A + \Sigma dx/\Sigma f$

(d) $\bar{X} = A + \dfrac{\Sigma f\, dx}{\Sigma f}$

Q. 66 Rs. 600 per day are paid on a research farm to its 50 daily paid labourers. A worker gets five unpaid holidays in a month. The average income of a daily paid labourer is:
(a) Rs. 250 p.m
(b) Rs. 300 p.m
(c) Rs. 350 p.m
(d) Rs. 400 p.m

Q. 67 The variate values which divide a series (frequency distribution) into five equal parts are called:
(a) quintiles
(b) quartiles
(c) octiles
(d) percentiles

Q. 68 The variate values which divide a series (frequency distribution) into four equal parts are called:
(a) quintiles
(b) quartiles
(c) deciles
(d) percentiles

Q. 69 The variate values which divide a series (frequency distribution) into ten equal parts are called:
(a) quartiles
(b) deciles
(c) octiles
(d) percentiles

Q. 70 The variate values which divide a series (frequency distribution) into 100 equal parts are known as:
(a) octiles
(b) quartiles
(c) percentiles
(d) deciles

Q. 71 The variate values which divide a series (frequency distribution) into eight equal parts are known as:
(a) quartiles
(b) quintiles
(c) deciles
(d) octiles

Q. 72 The number of partition values in case of quartiles is:
(a) 4
(b) 3
(c) 2
(d) 1

Q. 73 The number of partition values in case of quintiles cannot exceed:
(a) 4
(b) 3
(c) 2
(d) 1

Q. 74 For deciles, the total number of partition values are:
(a) 5
(b) 8
(c) 9
(d) 10

Q. 75 For percentiles, the total number of partition values are:
(a) 10
(b) 59
(c) 100
(d) 99

Q. 76 The second decile divides the series in the ratio:
(a) 1:1
(b) 1:2
(c) 1 : 4
(d) 2 : 5

Q. 77 Eightieth percentile divides a frequency distribution in the ratio:
(a) 4 : 1
(b) 4 : 5
(c) 3 : 1
(d) 3 : 2

Q. 78 The first quartile divides a frequency distribution in the ratio:
(a) 4 : 1
(b) 1 : 4
(c) 3 : 1
(d) 1 : 3

Q. 79 Fourth octile divides a frequency distribution in the ratio:
(a) 1 : 1
(b) 1 : 2
(c) 1 : 3
(d) 1 : 4

Q. 80 The first quartile is also known as:
(a) median
(b) lower quartile
(c) mode
(d) third decile

Q. 81 The third quartile is also called:
(a) lower quartile
(b) median
(c) mode
(d) upper quartile

Q. 82 In case of weighted mean, the accuracy or utility of the mean:
(a) decreases
(b) increases
(c) is unaffected
(d) none of the above

Q. 83 In certain situations weighted mean and unweighted mean can:
(a) be equal
(b) never be equal
(c) both (a) and (b)
(d) neither (a) and nor (b)

Q. 84 Weighted mean is more accurate if we use:
(a) estimated weights
(b) arbitrary weights
(c) real weights
(d) any of the above

Q. 85 If .3, .5, .8, .7, and 1.5 are the respective weights of the values 10, 15, 20, 25 and 30, then the weighted mean is:
(a) 20.0
(b) 23.42
(c) 16.58
(d) none of the above

Q. 86 If for a discrete series, the assumed mean $A = 50$, $\sum f\, dx = 45$ for $dx = x - A$, $\sum f = 12$, then the mean of the series is:
(a) 46.25
(b) 7.92
(c) 49.17
(d) 53.75

Q. 87 The mean of the following discrete series (frequency distribution),
x: 7, 12, 16, 22, 25
f: 4, 5, 8, 3, 2
is:
(a) 16.40
(b) 15.09
(c) 20.80
(d) none of the three

Q. 88 If for an individual series, assumed mean, $A = 25$, $\Sigma\, dx = -21$ for $dx = X - A$ and $N = 7$ then the mean of the series is:
(a) 20
(b) 21
(c) 22
(d) 6.57

Q. 89 Given the following less than type frequency distribution of income per month,

Income (Rs.) less than	*No. of persons*
1500	100
1250	80
1000	70
750	55
500	32
250	12

the median class of income is:
(a) 750-1000

(b) 1000-1250
(c) 250-500
(d) 500-750

Q. 90 For the distribution given in question 89, the modal class is:
(a) 250-500
(b) 500-750
(c) 750-1000
(d) none of the above

Q. 91 For the distribution given in question 89, the upper quartile class is:
(a) 500-750
(b) 750-1000
(c) 1000-1250
(d) 1250-1500

Q. 92 For the distribution given in question 89, the sixth decile class is:
(a) 500-750
(b) 750-1000
(c) 1000-1250
(d) 1250-1500

Q. 93 For the distribution given in question 89, 30th percentile class is:
(a) 250-500
(b) 500-750
(c) 750-1000
(d) none of the above

Q. 94 Given the following frequency distribution of income of employees,

Income Rs./month	*No. of employees*
0-250	12
250-500	20
500-750	23
750-1,000	15
1,000-1,250	10
1,250-1,500	20

The median income of employees is:
(a) 625.00
(b) 760.00
(c) 695.65
(d) none of the above

Q. 95 For the frequency distribution given is question 94, the mode of the distribution is:
(a) 23
(b) 666.66
(c) 513.33
(d) 568.18

Q. 96 For the grouped data given in question 94, the third quartile is:
(a) 1,200
(b) 1,125
(c) 1,183.33
(d) none of the above

Q. 97 Sixth decile of the continuous frequency distribution given in question 94 is:
(a) 833.33
(b) 1,000.00
(c) 70.00
(d) none of the above

Q. 98 20th percentile of the grouped data given in question 94 is:
(a) 250
(b) 350
(c) 500
(d) 550

Q. 99 The more than type frequency distribution of age of the patients on a particular day in a hospital is as given below.

Age (years)	*No. of patients*
> 10	152
> 20	128
> 30	113
> 40	77
> 50	36
> 60	22
> 70	5
and up to 80	

the 50th percentile class is:
(a) 30-40
(b) 40-50
(c) 50-60
(d) none of the above

Q. 100 For the more than type distribution given in question 99, the modal class is:
(a) 30-40
(b) 40-50
(c) 50-60
(d) none of the above

Q. 101 For the more than type distribution given in question 99, the first quartile class is:
(a) 30-40
(b) 40-50
(c) 50-60
(d) none of the above

Q. 102 In the ascending series of question 99, fourth decile class is:
(a) 30-40
(b) 40-50
(c) 50-60
(d) none of the above

Q. 103 The mode of the more than type distribution given in question 99 is:
(a) 37.78
(b) 41.56
(c) 42.56
(d) none of the above

Q. 104 The median (second quartile) of the more than type distribution given in question 99 is:
(a) 37.33
(b) 40.00
(c) 40.24
(d) 50.00

Q. 105 The seventh decile of the cumulative distribution of question 99 is:
(a) 42.34
(b) 48.78
(c) 57.66
(d) 47.66

Q. 106 Thirtieth percentile of the more than type distribution of question 99 is:
(a) 31.83
(b) 53.50
(c) 30.88
(d) none of the above

Q. 107 The upper quartile of the distribution given in question 99 is:
(a) 50.00
(b) 49.51
(c) 45.00
(d) 41.00

Q. 108 If we plot the more than type and less than type frequency distributions of the same set of data, their graphs intersect at the point which is known as:
(a) median
(b) mode
(c) mean
(d) none of the above

Q. 109 Mean of a set of values is based on:
(a) all values
(b) 50 per cent values
(c) first and last value
(d) maximum and minimum value

Q. 110 Which mean is most affected by extreme values?
(a) Geometric mean
(b) Harmonic mean
(c) Arithmetic mean
(d) Trimmed mean

Q. 111 The measure of central value which cannot be calculated with open end classes in case of grouped data is:
(a) median
(b) arithmetic mean
(c) mode
(d) third quartile

Q. 112 Harmonic mean gives more weightage to:
(a) small values
(b) large values
(c) positive values
(d) negative values

Q. 113 Harmonic mean gives less weightage to:
(a) positive values
(b) negative values
(c) small values
(d) large values

Q. 114 For further algebraic treatment geometric mean is:
(a) suitable
(b) not suitable
(c) sometimes suitable
(d) none of the above

Q. 115 For further algebraic treatment harmonic mean is:
(a) suitable

(b) not suitable
(c) sometimes suitable
(d) none of the above

Q. 116 For a highly variable series, the most suitable mean is:
(a) arithmetic mean
(b) geometric mean
(c) harmonic mean
(d) none of the above

Q. 117 The percentage of values in a set of values which are less than (more than) the median value is:
(a) 100 per cent
(b) 75 per cent
(c) 50 per cent
(d) 25 per cent

Q. 118 The percentage of values of a set which is beyond the third quartile is:
(a) 100 per cent
(b) 75 per cent
(c) 50 per cent
(d) 25 per cent

Q. 119 The percentage of data of a set which are to the left of seventh decile is:
(a) 30 per cent
(b) 50 per cent
(c) 70 per cent
(d) 90 per cent

Q. 120 The percentage of data of a set which is to the right of ninetieth percentile is:
(a) 0 per cent
(b) 10 per cent
(c) 90 per cent
(d) 80 per cent

Q. 121 In a class test, 40 students out of 50 passed with mean marks 6.0 and the overall average of class marks was 5.5. The average marks of students who failed were:
(a) 2.5
(b) 3.0
(c) 4.8
(d) 3.5

Q. 122 Seven persons gambled sitting on a table. Four persons lost on an average Rs. 55 whereas the other three gained on an average Rs. 70. Is the information worth believing?
(a) Yes
(b) No
(c) Not certain
(d) None of the above

Q. 123 The average marks of section A are 65 and that of section B are 70. The average of both the sections combined is 67. The ratio of number of students of section A to B is:
(a) 1 : 3
(b) 2 : 3
(c) 3 : 1
(d) 3 : 2

Q. 124 The partition value which divide a series into two equal parts is known as:
(a) second quartile
(b) third quintile
(c) fourth octile
(d) sixth deciles

Q. 125 In a distribution, the value around which the items tend to be most heavily concentrated is called:
(a) mean
(b) median
(c) third quartile
(d) mode

Q. 126 Geometric mean can be used to find out:
(a) population growth
(b) growth rate of GNP
(c) both the above
(d) none of (a) and (b)

Q. 127 Formula for the combined geometric mean 'GM_{12}' of two series of sizes n_1 and n_2 and their G.M.'s GM_1 and GM_2 respectively is:

(a) $GM_{12} = \dfrac{n_1 \log GM_1 + n_2 \log GM_2}{n_1 + n_2}$

(b) $GM_{12} = \dfrac{n_1 \times GM_1 + n_2 \times GM_2}{n_1 + n_2}$

(c) $GM_{12} = \dfrac{GM_1 \log n_1 + GM_2 \log n_2}{n_1 + n_2}$

(d) $GM_{12} = \text{Antilog}\left(\dfrac{n_1 \log GM_1 + n_2 \log GM_2}{n_1 + n_2}\right)$

Q. 128 Sum of the deviations about mean is:
(a) zero
(b) minimum
(c) maximum
(d) one

Q. 129 Sum of the absolute deviations about median is:
(a) zero
(b) maximum
(c) minimum
(d) one

Q. 130 Sum of square of the deviations about mean is:
(a) maximum
(b) minimum
(c) zero
(d) none of the above

Q. 131 Histogram is useful to determine graphically the value of:
(a) mean
(b) median
(c) mode
(d) all the above

Q. 132 Graphically partition values can be determined with the help of:
(a) frequency polygon
(b) bar diagram
(c) line diagram
(d) ogive curve

Q. 133 The suitable measure of central tendency for qualitative data is:
(a) mode
(b) arithmetic mean
(c) geometric mean
(d) median

Q. 134 Two series having the same mean, median and mode may:
(a) have same values
(b) not have same values
(c) both (a) and (b)
(d) none of (a) and (b)

Q. 135 Weighted mean gives a higher value than unweighted mean if:
(a) all the items have equal weights
(b) larger items have higher weights and small items have lower weights.
(c) larger items have lower weights and smaller items have higher weights
(d) none of the above

Q. 136 In a frequency distribution with open ends, one cannot find out:
(a) mean
(b) median
(c) mode
(d) all the above

Q. 137 The average of n observations $x_1, x_2, ..., x_n$ is M. If x_1 is replaced by x', then the new average is:

(a) $M - x_1 + x'$

(b) $\dfrac{M - x_1 + x'}{n}$

(c) $\dfrac{(n-1)M - x_1 + x'}{n}$

(d) $\dfrac{nM - x_1 + x'}{n}$

Q. 138 In case of 15 per cent trimmed mean, the percentage of observations utilised is:
(a) 15 per cent
(b) 30 per cent
(c) 85 per cent
(d) none of the above

Q. 139 If for values of X, A.M. = 25, H.M. = 9, then the G.M. is:
(a) 17
(b) 15
(c) 5.83
(d) 16

Q. 140 The second quartile of the following set of data, 0, 1, −1, −2, 6, 4, 5, 8, 12, 10, 11 is:
(a) 4
(b) 5
(c) 6
(d) 8

Q. 141 The measure of central tendency which remains unaltered by extreme observations is:
(a) mean
(b) mode
(c) harmonic mean
(d) geometric mean

Q. 142 The mean of the squares of first eleven natural numbers is:
(a) 46
(b) 23
(c) 48
(d) 42

Q. 143 The point of intersection of two cumulative frequency curves provides:
(a) mean
(b) mode
(c) median
(d) first quartile

Q. 144 The percentage of items in a frequency distribution lying between upper and lower quartiles is:
(a) 80 per cent
(b) 40 per cent
(c) 50 per cent
(d) 25 per cent

Q. 145 A person has been deputed to find the average income of factory employees. To provide a correct picture of average income, he should find out:
(a) geometric mean
(b) weighted mean
(c) progressive mean
(d) arithmetic mean.

Q. 146 To know the trend of a time series data, preferable type of average is:
(a) progressive average
(b) moving average
(c) weighted mean
(d) quadratic mean.

Q. 147 If we know the mean values and number of units of various groups, then one mean value for all the groups can be obtained by finding out:
(a) geometric mean
(b) weighted mean
(c) pooled mean
(d) progressive mean

Q. 148 In a frequency distribution of a large number of values, the mode is:
(a) largest observation
(b) smallest value
(c) observation with maximum frequency
(d) maximum frequency of an observation

Q. 149 The relation between quadratic mean (Q.M.) and arithmetic mean (A.M.) is:
(a) Q.M. = A.M.
(b) Q.M. > A.M.
(c) Q.M. < A.M.
(d) Q.M. ≠ A.M.

Q. 150 A set values contains a few negative values. The average of set can better be measured by:
(a) arithmetic mean
(b) geometric mean
(c) progressive mean
(d) quadratic mean

ANSWERS

Section-B

(1) John I. Griffin (2) R.A. Fisher (3) A.L. Bowley (4) typical value, summary measure (5) extreme values (6) variate (7) summarises (8) mean (9) 5 (10) $\frac{1}{10}$th (11) 13 (12) 16 (13) zero (14) Rs. 65 (15) Rs. 27 (16) accurate (17) 70 (18) $(n + 1)/2$ (19) minimum (20) 8 (21) Antilog (G_1/G_2) (22) zero (23) ratio; proportions (24) 4 (25) 5 (26) average speed (27) infinity (28) G.M. = $\sqrt{\text{A.M.} \times \text{H.M.}}$ (29) median (30) minimum (31) 12 (32) unique (33) median (34) open end (35) mode (36) 7 (37) median; mode (38) mode (39) mode (40) unimodal (41) bimodal (42) grouping (43) quartile (44) median (45) 75th (46) fifth (47) D_7 (48) 26 per cent (49) lower or first (50) 60th (51) 50 per cent (52) 34 per cent (53) 40th (54) $Q_2 < D_6 < P_{80}$ (55) fifth (56) fiftieth (57) ascending (58) quintiles (59) P_{80} (60) third (61) first; twenty-fifth (62) fractiles

(63) Ya Lun Chou (64) $Q_1 = P_{25}$ (65) $D_4 < P_{45}$ (66) 75 per cent (67) numerical; analytical (68) $(Q_1 + Q_3)/2$ (69) antilog (70) reciprocal (71) negative (72) H.M. ≤ G.M. ≤ A.M. (73) same (74) standard deviation (75) 26 (76) 7 (77) 21 (78) entire data (79) best (80) not rigidly.

SECTION-C

(1) a	(2) a	(3) b	(4) b	(5) a	(6) b
(7) b	(8) c	(9) c	(10) d	(11) c	(12) a
(13) b	(14) c	(15) a	(16) d	(17) b	(18) a
(19) b	(20) c	(21) c	(22) d	(23) b	(24) c
(25) d	(26) a	(27) c	(28) a	(29) a	(30) c
(31) a	(32) d	(33) c	(34) a	(35) d	(36) d
(37) c	(38) b	(39) c	(40) b	(41) c	(42) d
(43) d	(44) c	(45) d	(46) a	(47) c	(48) b
(49) a	(50) b	(51) c	(52) b	(53) a	(54) d
(55) b	(56) d	(57) d	(58) d	(59) a	(60) d
(61) d	(62) a	(63) c	(64) b	(65) d	(66) b
(67) a	(68) b	(69) b	(70) c	(71) d	(72) b
(73) a	(74) c	(75) d	(76) c	(77) a	(78) d
(79) a	(80) b	(81) d	(82) b	(83) a	(84) c
(85) b	(86) d	(87) b	(88) c	(89) d	(90) b
(91) c	(92) b	(93) a	(94) c	(95) d	(96) b
(97) a	(98) b	(99) b	(100) b	(101) d	(102) a
(103) d	(104) c	(105) d	(106) a	(107) b	(108) a
(109) a	(110) c	(111) b	(112) a	(113) d	(114) b
(115) b	(116) c	(117) c	(118) d	(119) c	(120) b
(121) d	(122) b	(123) d	(124) a	(125) d	(126) c
(127) d	(128) a	(129) c	(130) b	(131) c	(132) d
(133) d	(134) c	(135) b	(136) a	(137) d	(138) d
(139) b	(140) b	(141) b	(142) a	(143) c	(144) c
(145) b	(146) b	(147) c	(148) c	(149) b	(150) d

Suggested Reading

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107. Standard deviation gives more weight to __________ values.

108. Standard deviation is useful in finding the descriptive measures like __________ and __________ of a distribution.

109. Variation in data can graphically be perceived by __________.

110. If the sum of deviations from median is not zero, then a distribution will be __________.

111. If the frequencies on either side of mode are not similarly distributed, the frequency distribution curve will be __________.

112. Measure of skewness provides the __________ and __________ of asymmetry present in a distribution.

113. Measure of Kurtosis shows the degree of __________ of a frequency distribution curve.

114. Standard deviation of two values X_1 and X_2 is equal to __________.

115. Change of origin and scale of values for a set makes the calculation of standard deviation __________.

SECTION-C

Multiple Choice Questions

Select the correct alternative out of given ones:

Q. 1 Which of the following is not a measure of dispersion?
(a) mean deviation
(b) quartile deviation
(c) standard deviation
(d) average deviation from mean

Q. 2. Which of the following is a unitless measure of dispersion?
(a) standard deviation
(b) mean deviation
(c) coefficient of variation
(d) range

Q. 3 Which one of the given measures of dispersion is considered best?
(a) standard deviation
(b) range
(c) variance
(d) coefficient of variation

Q. 4. For comparison of two different series, the best measure of dispersion is:
(a) range
(b) mean deviation
(c) standard deviation
(d) none of the above

Q. 5. Correct formula for mean deviation from a constant A of a series in which the variate values $x_1, x_2, ..., x_k$ have frequencies $f_1, f_2, ..., f_k$ respectively is:

(a) $\frac{1}{N}\sum_i (f_i x_i - A)$

(b) $\frac{1}{N}\sum_i f_i (x_i - A)$

(c) $\frac{1}{N}\sum_i |f_i (x_i - A)|$

(d) $\frac{1}{N}\sum_i |f_i x_i - \bar{A}|$

where $i = 1, 2, ..., k$ and $\sum_i f_i = N$

Q. 6. Correct formula for variance of n sample observations $x_1, x_2, ..., x_n$ is:

(a) $\frac{1}{n-1}\sum_i (x_i - \bar{x})^2$

(b) $\frac{1}{n-1}\left(\sum_i x_i^2 - \bar{x}^2\right)$

(c) $\frac{1}{n}\sum_i (x_i - \bar{x})^2$

(d) $\frac{1}{n}\sum_i x_i^2 - \bar{x}^2$

Q. 7 The correct relation between variance and standard deviation (S.D.) of a variable X is:
(a) S.D. = $[\text{Var}(X)]^2$
(b) S.D. = $[\text{Var}(X)]^{1/2}$
(c) S.D. = Var (X)
(d) none of the above

Q. 8 Formula for coefficient of variation is:

(a) $C.V. = \frac{S.D.}{\text{mean}} \times 100$

(b) $C.V. = \frac{\text{mean}}{S.D.} \times 100$

(c) $C.V. = \frac{\text{mean} \times S.D.}{100}$

(d) $C.V. = \frac{100}{\text{mean} \times S.D.}$

Q. 9 Out of all measures of dispersion, the easiest one to calculate is:
(a) standard deviation
(b) range
(c) variance
(d) quartile deviation

Q. 10 Formula for range (R) of a set of values $X_1, X_2, ..., X_n$ is:

(a) $R = X_{min} - X_{max}$

(b) $R = |X_{min} - X_{max}|$

(c) $R = X_{max} - X_{min}$

(d) $R = X_n - X_1$

Q. 11 Formula for coefficient of range of the set of observations $X_1, X_2, ..., X_n$ is:

(a) coeff. of range $= \frac{X_{max} - X_{min}}{X_{max} + X_{min}}$

(b) coeff. of range $= \frac{X_{max} + X_{min}}{X_{max} - X_{min}}$

(c) coeff. of range $= \frac{X_{max}}{X_{min}}$

(d) coeff. of range $= \frac{X_{max} - X_{min}}{X_{max}}$

Q. 12 Coefficient of quartile deviation is given by the formula:

(a) coeff. of $Q.D. = \frac{Q_3 + Q_1}{Q_3 - Q_1}$

(b) coeff. of $Q.D. = \frac{Q_3 + Q_1}{Q_1 - Q_3}$

(c) coeff. of $Q.D. = \frac{Q_3 - Q_1}{Q_1 - Q_3}$

(d) coeff. of $Q.D. = \frac{Q_3 - Q_1}{Q_3 + Q_1}$

Q. 13 For a symmetrical distribution, $M_d \pm Q.D.$ covers:
(a) 25 per cent of the observations
(b) 50 per cent of the observations
(c) 75 per cent of the observations
(d) 100 per cent of the observations
M_d = Median and Q.D. = Quartile deviation

Q. 14 Quartile deviation or semi inter-quartile deviation is given by the formula:

(a) $Q.D. = \frac{Q_3 + Q_1}{2}$

(b) $Q.D. = Q_3 - Q_1$

(c) $Q.D. = (Q_3 - Q_1)/2$

(d) $Q.D. = (Q_3 - Q_1)/4$

Q. 15 Mean deviation is minimum when deviations are taken from:
(a) mean
(b) median
(c) mode
(d) zero

Q. 16 Sum of squares of the deviations is minimum when deviations are taken from:
(a) mean
(b) median
(c) mode
(d) zero

Q. 17 If a constant value 5 is subtracted from each observation of a set, the variance is:

(a) reduced by 5
(b) reduced by 25
(c) unaltered
(d) increased by 25

Q. 18 If each observation of a set is divided by 10, the S.D. of the new observations is:

(a) $\frac{1}{10}$th of S.D. of original obs
(b) $\frac{1}{100}$th of S.D. of original obs
(c) not changed
(d) 10 times of S.D. of original obs

Q. 19 If each value of a set is divided by c, then the variance (s^2) of the original sample values from coded values $dx_1, dx_2, \ldots, dx_n$ is obtained by the formula:

(a) $s^2 = \frac{1}{n-1}\sum_i \left(dx_i - \bar{d}x\right)^2 \times c$

(b) $s^2 = \frac{1}{n-1}\sum_i \left(dx_i - \bar{d}x\right)^2 \times c^2$

(c) $s^2 = \frac{1}{n-1}\sum_i \left(dx_i - \bar{d}x\right)^2 \times \frac{1}{c}$

(d) $s^2 = \frac{1}{n-1}\sum_i \left(dx_i - \bar{d}x\right)^2 \times \frac{1}{c^2}$

Q. 20 Which of the following formula for standard deviation of a frequency distribution is not correct?

(a) $\sigma = \sqrt{\frac{1}{N}\sum_i f_i(x_i - \bar{x})^2}$

(b) $\sigma = \sqrt{\frac{1}{N}\sum_i f_i x_i^2 - \bar{x}^2}$

(c) $\sigma = \sqrt{\frac{1}{N}\sum_i f_i x_i^2 - \left(\frac{\Sigma_i f_i x_i}{N}\right)^2}$

(d) $\sigma = \sqrt{\frac{1}{N}\left(\sum_i f_i x_i\right)^2 - \frac{\Sigma_i f_i x_i}{N}}$

where $\Sigma_i f_i = N$ and other notations are as usual.

Q. 21 If a constant 'c' is subtracted from each observation and then divided by d, then the formula for variance of frequency distribution having k groups is:

(a) $\sigma^2 = \left[\frac{1}{N}\sum_{i=1}^{k} f_i x_i'^2 - \left(\frac{\sum_{i=1}^{k} f_i x_i'}{N}\right)^2\right] \times d^2$

(b) $\sigma^2 = \frac{1}{N-1}\left[\sum_{i=1}^{k} f_i (x_i' - \bar{x}')^2\right] \times d^2$

(c) $\sigma^2 = \left[\frac{1}{N}\sum_{i=1}^{k} f_i x_i'^2 - \frac{\left(\sum_{i=1}^{k} f_i x_i'\right)^2}{N}\right] \times d^2$

(d) None of the above

where $x_i' = \frac{x_i - c}{d}$ and $N = \sum_i f_i$

Q. 22 The empirical relationship between quartile deviation (Q.D.) and standard deviation in normal distribution is:

(a) 3 Q.D. $\doteq$ 2 S.D.
(b) 4 Q.D. $\doteq$ 3 S.D.
(c) 6 Q.D. $\doteq$ 5 S.D.
(d) 5 Q.D. $\doteq$ 4 S.D.

Q. 23 The relationship between mean deviation (M.D.) and standard deviation is:

(a) 3 M.D. = 2 S.D.
(b) 5 M.D. = 4 S.D.
(c) 6 M.D. = 5 S.D.
(d) M.D. = S.D.

Q. 24 The empirical relation between quartile deviation (Q.D.) and mean deviation (M.D.) from mean is:

(a) 3 Q.D. $\doteq$ 5 M.D.
(b) 6 Q.D. $\doteq$ 3 M.D
(c) 5 Q.D. $\doteq$ 6 M.D.
(d) 6 Q.D. $\doteq$ 5 M.D.

Q. 25 An empirical relation between standard deviation, mean deviation about mean and quartile deviation is:

(a) 4 S.D. $\doteq$ 6 M.D. $\doteq$ 5 Q.D.
(b) 4 S.D. $\doteq$ 5 M.D. $\doteq$ 6 Q.D.
(c) 6 S.D. $\doteq$ 5 M.D. $\doteq$ 4 S.D.
(d) 5 S.D. $\doteq$ 4 M.D. $\doteq$ 6 Q.D.

Q. 26 The empirical relation between range (R) and standard deviation is:

(a) R = 3 S.D.
(b) R = 2 S.D.
(c) R = 6 S.D.
(d) R = 4 S.D.

Q. 27 An empirical relationship between range and quartile deviation about mean is:

(a) R = 4 Q.D.
(b) R = 9 Q.D.
(c) R = 6 Q.D.
(d) none of the above

Q. 28 An empirical relation between range and mean deviation is:

(a) R = 10 M.D.
(b) $2R$ = 5 M.D.
(c) $3R$ = 5 M.D.
(d) $2R$ = 15 M.D.

Q. 29 Which measure of dispersion ensures highest degree of reliability?

(a) range
(b) mean deviation
(c) quartile deviation
(d) standard deviation

Q. 30 Which measure of dispersion ensures lowest degree of reliability?

(a) range
(b) mean deviation
(c) quartile deviation
(d) standard deviation

Q. 31 There are two populations consisting of N_1 and N_2 units, having means $\bar{X}_1$ and $\bar{X}_2$, variances σ_1^2 and σ_2^2 respectively. Let their pooled mean be $\bar{X}_{12}$. Also, supposing $\bar{X}_1 - \bar{X}_{12} = d_1$ and $\bar{X}_2 - \bar{X}_{12} = d_2$. The formula for pooled variance is:

(a) $\sigma^2 = \frac{N_1\left(\sigma_1^2 + d_1^2\right) + N_2\left(\sigma_2^2 + d_2^2\right)}{N_1 + N_2}$

(b) $\sigma^2 = \frac{N_2\left(\sigma_1^2 + d_1^2\right) + N_1\left(\sigma_2^2 + d_2^2\right)}{N_1 + N_2}$

(c) $\sigma^2 = \frac{N_1\sigma_1^2 + N_2 d_1^2 + N_2\sigma_2^2 + N_1 d_2^2}{N_1 + N_2}$

(d) $\sigma_2 = \frac{N_2\sigma_1^2 + N_1\sigma_2^2 + N_1 d_1^2 + N_2 d_2^2}{N_1 + N_2}$

Q. 32 Average wages of workers of a factory are Rs. 550.00 per month and the standard deviation of wages is 110. The coefficient of variation is:

(a) C.V. = 30 per cent
(b) C.V. = 15 per cent
(c) C.V. = 500 per cent
(d) C.V. = 20 per cent

Q. 33 If the mean deviation of a distribution is 20.20, the standard deviation of the distribution is:

(a) 15.15
(b) 25.25
(c) 30.30
(d) none of the above

Q. 34 If the mean and standard deviation of A and B are as, $\bar{X}_A = 15.0$, $\bar{X}_B = 20.0$ and $\sigma_A^2 = 25$ and $\sigma_B^2 = 16$, which of the two series is more consistent.

(a) series A
(b) series B
(c) series A and B are equally consistent
(d) none of the above

Q. 35 If the standard deviation of a distribution is 15, the quartile deviation of the distribution is:

(a) 15.0
(b) 12.5
(c) 10.0
(d) none of the above

Q. 36 If the quartile deviation of a series is 60, the mean deviation of this series is:
(a) 72
(b) 48
(c) 50
(d) 75

Q. 37 In a discrete set of values, the correct relation between mean deviation and standard deviation is:
(a) M.D. > S.D.
(b) M.D. < S.D.
(c) M.D. ≤ S.D.
(d) M.D. ≥ S.D.

Q. 38 If the first 25 per cent observations of a series are 20 or less and last 25 per cent observations of a series are 50 or more, the quartile deviation (semi inter-quartile deviation) is:
(a) 25
(b) 35
(c) 15
(d) 30

Q. 39 The mean and standard deviation of a set of values are 25 and 5, respectively. If a constant value 5 is added to each value, the coefficient of variation of the new set of values is:
(a) 250 per cent
(b) 600 per cent
(c) 20 per cent
(d) 16.6 per cent

Q. 40 If the mean of a series is 10 and its coefficient of variation is 40 per cent, the variance of the series is:
(a) 4
(b) 8
(c) 12
(d) none of the above

Q. 41 which of the following measures of dispersion can attain a negative value?
(a) range
(b) mean deviation
(c) standard deviation
(d) variance

Q. 42 A set of values is said to be relatively uniform if it has:
(a) high dispersion
(b) zero dispersion
(c) little dispersion
(d) negative dispersion

Q. 43 The mean and standard deviation of a set of values from a normal distribution are 66 and 4, respectively. The range in which almost 95 per cent values lie is:
(a) 62 to 70
(b) 62 to 74
(c) 58 to 74
(d) 66 and 74

Q. 44 The measure of dispersion which ignores signs of the deviations from a central value is:
(a) range
(b) quartile deviation
(c) standard deviation
(d) mean deviation

Q. 45 Which measure of dispersion is least affected by extreme values?
(a) range
(b) mean deviation
(c) standard deviation
(d) quartile deviation

Q. 46 Which measure of dispersion is most affected by extreme values?
(a) range
(b) mean deviation
(c) standard deviation
(d) quartile deviation

Q. 47 Range of a set of values is 65 and maximum value in the series is 83. The minimum value of the series is:
(a) 74
(b) 9
(c) 18
(d) none of the above

Q. 48 If the minimum value in a set is 9 and its range is 57, the maximum value of the set is
(a) 33
(b) 66
(c) 48
(d) none of the above

Q. 49 If the values of a set are measured in cm, the unit of variance will be:
(a) no unit
(b) cm
(c) cm^2
(d) cm^3

Q. 50 Which measure of dispersion has a different unit other than the unit of measurement of values:
(a) range
(b) mean deviation
(c) standard deviation
(d) variance

Q. 51 The average of the sum of squares of the deviations about mean is called:
(a) variance
(b) absolute deviation
(c) standard deviation
(d) mean deviation

Q. 52 Quartile deviation is equal to:
(a) interquartile range
(b) double the interquartile range
(c) half of the interquartile range
(d) none of the above

Q. 53 Which measure of dispersion can be calculated in case of open end intervals?
(a) range
(b) standard deviation
(c) coefficient of variation
(d) quartile deviation

Q. 54 Which one property out of the following does not hold good in case of standard deviation?
(a) It is distorted by extreme values
(b) It is not very sensitive to sampling fluctuations as compared to other measures.
(c) It is a unitless measure of dispersion
(d) It is a most used measure of dispersion

Q. 55 If each value of a series is divided by 5, its coefficient of variation is reduced by:
(a) 0 per cent
(b) 5 per cent
(c) 10 per cent
(d) 20 per cent

Q. 56 If each value of a series is multiplied by 10, the coefficient of variation will be increased by:
(a) 5 per cent
(b) 10 per cent
(c) 15 per cent
(d) 0 per cent

Q. 57 If a constant value 10 is subtracted from each value of a series, the coefficient of variation will be:
(a) decreased in comparison to original value
(b) increased in comparison to original value
(c) same as original value
(d) none of the above

Q. 58 If each value of a series is multiplied by a constant 'c', the coefficient of variation as compared to original value is:
(a) increased
(b) decreased
(c) unaltered
(d) zero

Q. 59 If each value of a set is divided by a constant 'd', the coefficient of variation will be:
(a) same as original value
(b) less than original value
(c) more than original value
(d) none of the above

Q. 60 For a positive skewed distribution, which of the following inequality holds?
(a) median > mode
(b) mode > mean
(c) mean > median
(d) mean > mode

Q. 61 For a negatively skewed distribution, the correct inequality is:
(a) mode < median
(b) mean < median

(c) mean < mode
(d) none of the above

Q. 62 For a positive skewed frequency curve, the inequality that holds is:
(a) $Q_1 + Q_3 > 2Q_2$
(b) $Q_1 + Q_2 > 2Q_3$
(c) $Q_1 + Q_3 > Q_2$
(d) $Q_3 - Q_1 > Q_2$

Q. 63 For a negatively skewed frequency distribution curve, the third central moment,
(a) $\mu_3 > 0$
(b) $\mu_3 < 0$
(c) $\mu_3 = 0$
(d) μ_3 does not exist

Q. 64 For a symmetrical distribution, the coefficient of skewness:
(a) $\alpha_3 = 1$
(b) $\alpha_3 = 3$
(c) $\alpha_3 = 0$
(d) $\alpha_3 = -1$

Q. 65 For a leptokurtic frequency curve, the measure of Kurtosis,
(a) $\alpha_4 = 0$
(b) $\alpha_4 = -3$
(c) $\alpha_4 < 1$
(d) $\alpha_4 > 3$

Q. 66 In case of a positive skewed distribution, the relation between mean, median and mode that holds is:
(a) median > mean > mode
(b) mean > median > mode
(c) mean = median = mode
(d) none of the above

Q. 67 If a moderately skewed distribution has mean 30 and mode 36, the median of the distribution is:
(a) 30
(b) 28
(c) 32
(d) none of the above

Q. 68 If a moderately skewed distribution has mean 40 and median equal to 30, the mode of the distribution is:
(a) 10
(b) 33
(c) 20
(d) zero

Q. 69 First and third quartiles of a frequency distribution are 30 and 75. Also its coefficient of skewness is 0.6. The median of the frequency distribution is:
(a) 40
(b) 39
(c) 38
(d) 41

Q. 70 If the mean, standard deviation and coefficient of skewness of a frequency distribution are 60, 45 and –0.4, respectively, the mode of the frequency distribution is:
(a) 80
(b) 82
(c) 78
(d) 68

Q. 71 For a moderately skew distribution, the empirical relation between mean (M), median (M_d) and mode (M_0) is:
(a) $3(M - M_0) = M - M_d$
(b) $3(M_d - M) = M_0 - M$
(c) $3(M - M_d) = M - M_0$
(d) $2(M_0 - M) = 3(M_d - M)$

Q. 72 If the first quartile $Q_1 = 15$ and third quartile $Q_3 = 25$, the coefficient of quartile deviation is:
(a) 4
(b) 1/4
(c) 5/3
(d) 3/5

Q. 73 If the first quartile $Q_1 = 20$ and third quartile $Q_3 = 50$, the quartile deviation is:
(a) 35
(b) 15
(c) 2.5
(d) 0.8

Q. 74 For a negatively skewed distribution, the correct relation between mean, median and mode is:

(a) mean = median = mode
(b) median < mean < mode
(c) mean < median < mode
(d) mode < mean < median

Q. 75 If the mode of a frequency distribution $M_0 = 16$ and its mean $\overline{X} = 16$, the median of the distribution is:
(a) zero
(b) 16
(c) 32
(d) 8

Q. 76 In case of positive skewed distribution, the extreme values lie in the
(a) left tail
(b) right tail
(c) middle
(d) anywhere

Q. 77 The extreme values in a negatively skewed distribution lie in the:
(a) middle
(b) right tail
(c) left tail
(d) whole curve

Q. 78 The relation between variance and standard deviation is:
(a) variance is the square root of standard deviation
(b) standard deviation is the square of the variance
(c) variance is equal to standard deviation
(d) square of the standard deviation is equal to variance

Q. 79 Which of the following statements is true for a measure of dispersion?
(a) mean deviation does not follow algebraic rule
(b) range is a crudest measure
(c) coefficient of variation is a relative measure
(d) all the above statements

Q. 80 For a set of values:
(a) mean deviation is always less than standard deviation
(b) mean deviation is always greater than standard deviation
(c) mean deviation is always equal to standard deviation
(d) none of the above

Q. 81 Variance of the following frequency distribution,

Classes	*Frequency*
2-4	2
4-6	5
6-8	4
8-10	1

is approximately equal to:
(a) 2.5
(b) 2.9
(c) 5.0
(d) none of the above

Q. 82 For the data given in question 90 mean deviations about median is:
(a) 1.43
(b) 1.00
(c) 2.43
(d) 6

Q. 83 Coefficient of variation for the data given in question 90 is:
(a) 48.33 per cent
(b) 206.90 per cent
(c) 195.17 per cent
(d) 30.03 per cent

Q. 84 The range of values for the frequency distribution given in question 81 is:
(a) 2
(b) 10
(c) 8
(d) 6

Q. 85 For the data given in question 81, the coefficient of quartile deviation is:
(a) 4.385
(b) 0.228
(c) 2.6
(d) 11.4

Q. 86 The range of the set of values, 15, 12, 27, ... 9, 18, 21 is:
(a) 21
(b) 4.5
(c) 0.64
(d) 3

Q. 87 The coefficient of range for the values given in question 86 is:
(a) 1.571
(b) 4.500
(c) 0.636
(d) 0.222

Q. 88 The coefficient of skewness of a series A is 0.15 and that of series B 0.062. Which of the two series is less skew?
(a) series A
(b) series B
(c) no decision
(d) none of the above

Q. 89 If the coefficient of Kurtosis γ_2 of a distribution is zero, the frequency curve is:
(a) leptokurtic
(b) platykurtic
(c) mesokurtic
(d) any of the above

Q. 90 If for a distribution, coefficient of Kurtosis $\gamma_2 < 0$, the frequency curve is:
(a) leptokurtic
(b) platykurtic
(c) mesokurtic
(d) any of the above

Q. 91 The value of coefficient of Kurtosis β_2 can be:
(a) less than 3
(b) greater than 3
(c) equal to 3
(d) all the above

Q. 92 The standard deviation of a set of values will be:
(a) positive when the values are positive
(b) positive when the values are negative
(c) always positive
(d) all the above

Q. 93 Sum of square of the deviations is minimum when the deviations are taken from:
(a) mean
(b) median
(c) mode
(d) an arbitrary value.

Q. 94 The relationship between the variance of median [V (med)] and variance of mean [V (mean)] of a normal population is:
(a) V (med) < V (mean)
(b) V (med) = V (mean)
(c) V (med) > V (mean)
(d) 1.57 V (med) = V (mean)

Q. 95 Kurtosis in frequency distribution is adjudged around:
(a) second quartile
(b) arithmetic mean
(c) quadratic mean
(d) mode.

Q. 96 The variance of first n natural numbers is:
(a) $(n^2+1)/12$
(b) $(n+1)^2/12$
(c) $(n^2-1)/12$
(d) $(2n^2-1)/8$

Q. 97 If a random variable X has mean 3 and standard deviation 5, then the variance of a variable $Y = 2X - 5$ is:
(a) 45
(b) 100
(c) 15
(d) 40

Q. 98 Kurtosis and skewness of a frequency distribution curve are bound by the relation:
(a) they always coexist
(b) either of the two can exist alone
(c) their measures are always positive
(d) their measures are always negative.

Q. 99 All values in a sample are same. Then their variance is:
(a) zero
(b) one
(c) not calculable
(d) all the above

Q. 100 Calculation of pooled variance of two series of sizes n_1 and n_2 requires:
(a) means of individual series
(b) variances of individual series
(c) pooled mean
(d) all the above

ANSWERS

SECTION-B

(1) W.I. King (2) Reigleman (3) A.L. Bowley (4) B.C. Brookes and W.F.L. Dick (5) Speigel (6) C.T. Clark and L.L. Schkade (7) John I. Griffin (8) range (9) Range (10) relative (11) $Q_3 - Q_1$ (12) $P_{90} - P_{10}$ (13) $D_9 - D_1$ (14) all observations (15) median (16) positive (17) δ_c/c (18) $\Sigma f|X - M_d|/\Sigma f$ (19) square root (20) mean error; mean square error; root mean square deviation from mean (21) standard deviation (22) coefficient of variation (23) S.D. = $\sqrt{\text{variance}}$ (24) Lesser (25) Quartile (26) 50 (27) positional (28) symmetric (29) absolute (30) Range (31) reliability (32) twice (33) median (34) second (35) P.E. $= \frac{2}{3}\sigma$ (36) 5 : 6 (37) 3 : 2 (38) central (39) raw (40) zero (41) mid- points (42) zero (43) first (44) second (45) mode (46) equal (47) more (48) $\gamma_2 = \beta_2 - 3$ (49) $\mu_4 > 3\mu_2^2$ (50) three (51) less than zero (52) $3\mu_2^2 > \mu_4$ (53) not direction (54) bulginess (55) asymmetry (56) minimum (57) $P_{90} + P_{10} = 2P_{50}$ (58) $D_9 + D_1 = 2D_5$ (59) $\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$ (60) no (61) affects (62) Moments (63) standard deviation (64) (mean-mode)/S.D. (65) mesokurtic (66) leptokurtic (67) positive (68) positive (69) normal (70) mode (71) median (second quartile) (72) less than (73) mean − mode = 3 (mean − median) (74) negative (75) mean (76) percentage (77) Lorenz curve (78) cumulate (79) not same (80) 0.8 (81) 66.66 (82) 60 per cent (83) team B (84) second (85) first (86) less than (87) More (88) variance (89) average value (90) 25 (91) 20 (92) $\left(\frac{1}{n}\sum f|X - A|\right)/A$ (93) zero (94) greater than (95) 15 per cent (96) 0.25 (97) 37 (98) 21.66 (99) median (100) 4625 (101) $\pi\sigma^2/2n$ (102) quadratic (103) greater (104) A.M. ≤ S.D. (105) $\sqrt{(n^2$ [illegible] (106) mode (107) extreme (108) skewness; [illegible] (109) Lorenz curve (110) skewed (111) [illegible] asymmetrical (112) magnitude; direction (113) convexity (114) $(X_1 - X_2)/2$ (115) easier [illegible] simpler.

SECTION-C

(1) d	(2) c	(3) a	(4) d	(5) c	(6) [illegible]
(7) b	(8) a	(9) b	(10) c	(11) a	(12) [illegible]
(13) b	(14) c	(15) b	(16) a	(17) c	(18) [illegible]
(19) b	(20) d	(21) a	(22) a	(23) b	(24) [illegible]
(25) b	(26) c	(27) b	(28) d	(29) d	(30) [illegible]
(31) a	(32) d	(33) b	(34) b	(35) c	[illegible]
(37) b	(38) c	(39) d	(40) d	(41) a	(42) [illegible]
(43) c	(44) d	(45) d	(46) a	(47) c	(48) [illegible]
(49) c	(50) d	(51) a	(52) c	(53) d	(54) [illegible]
(55) a	(56) d	(57) b	(58) c	(59) a	(60) a
(61) c	(62) a	(63) b	(64) c	(65) d	(66) b
(67) c	(68) a	(69) b	(70) c	(71) [illegible]	(72) [illegible]
(73) b	(74) c	(75) b	(76) b	(77) c	(78) d
(79) d	(80) a	(81) b	(82) a	(83) d	(84) [illegible]
(85) b	(86) a	(87) c	(88) b	(89) c	(90) b
(91) d	(92) c	(93) a	(94) c	(95) d	(96) [illegible]
(97) b	(98) b	(99) a	(100) d		

Suggested Reading

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5. Hoel, P.G. and Jessen, R.J., *Basic Statistics for Business and Economics*, John Wiley, New York, 1982.